

Exercise 3 : Tries

Task 1: Verifying Correctness

[code in the wet part : `Figure`]

Result : the test come up with errors for

`nextwordInTrie(String word)` and `findNextwordFromNode(TrieNode, prefix)`

Task 2 : Bug Identification

1. In `findNextwordFromNode(TrieNode node, String prefix)`

It fails to **check** if the current node is the **end** of a valid word. This bug leads to it incorrectly skipping over valid words and jumps straight into the children.

Here's the test case reveals its failure:

```
Trie.TrieNode sNode = root.children['s' - 'a'];
assertEquals("ame", trie.findNextwordFromNode(sNode, "s"));
```

"same" exists in the trie and should be returned. However, `node.iswordEnd` isn't checked, the method skips "same" and dives into deeper children.

2. In `nextwordInTrie(String word)`

It tried to access the root if `j == -1`, but the loop starts at `path.size() - 2`, so `j` is never `-1`. As a result, it may skip valid alternatives.

Here's the test case reveals its failure:

```
assertEquals("save", trie.nextwordInTrie("same"));
```

The root can not be accessed, then the backtracking algorithm goes wrong. So it will never explore "save" from the parent node 's' after reaching "same".

Task 3: Bug Fixing

[code in the wet part : `Fix`]